

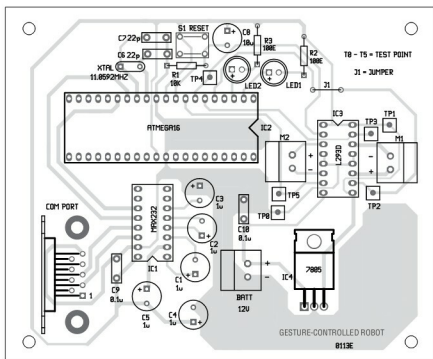


Fig. 8: Component layout for the PCB

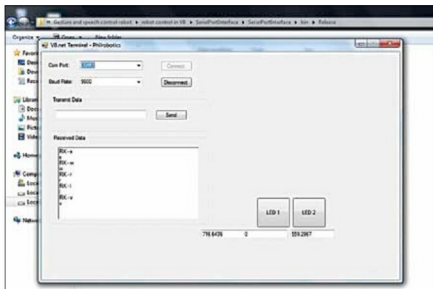


Fig. 9: VB application program

the components on the PCB and connect the motors and battery to build the robot. Use suitable bases for the ICs. Before inserting the MCU in the circuit, burn the program into it using a suitable programmer.

Connect the robot to the computer using the USB-to-serial converter as shown in the block diagram. Corresponding drivers for USB-to-serial converter may need to be installed.

Check whether the USB-to-serial converter is detected in the device manager and change the COM port to '2'.

Run application. Once the robot and Kinect sensor are properly interfaced with the computer, download the required software and run 'SerialPortInterface.exe' from the location "Gesture controlled robot\VB Application Program\SerialPortInterface\SerialPortInterface\bin\Release."

PARTS LIST

Semiconductors:

- IC1 - MAX232 RS interface
- IC2 - ATmega16 microcontroller
- IC3 - L293D motor driver
- IC4 - 7805, 5V regulator
- LED1, LED2 - 5mm LED

Resistors (all ¼-watt, ±5 per cent carbon):

- R1 - 10-kilo-ohm
- R2, R3 - 100-ohm

Capacitors:

- C1-C5 - 1µF, 16V electrolytic
- C6, C7 - 22pF ceramic disk
- C8 - 10µF, 16V electrolytic
- C9, C10 - 0.1µF ceramic disk

Miscellaneous:

- S1 - Push-to-on tactile switch
- X₁ - 11.0592MHz crystal
- M₁, M₂ - 12V, 100-rpm geared motor
- BATT - 12V, 4.5Ah battery
- COM port - DB9 female connector

EFY Note

The source codes of this project are included in this month's EFY DVD and are also available for free download on www.efymag.com website.

The application program is shown in Fig. 9.

The COM port and baud rate are automatically selected. Now the movement of the operator in front of Kinect Sensor can be detected with the changing coordinates at the bottom of the application program. All the registered gestures that an operator does in front of Kinect sensor are reflected back in the 'Received Data' section and sent to the COM port for the movement of the robot.

LED1 and LED2 buttons are for testing purpose. Once pressed, these will turn on the two LEDs by sending letters 'a' and 'b', respectively, to confirm the serial connectivity.

Check the correct power supply as 5V at TP1 with respect to TP0. Make some registered gesture in front of Kinect sensor and check corresponding logic inputs at TP2 through TP4. ●

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